Sahaj Bhadja

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Summary

Customer-focused AI engineer experienced in **Python, JavaScript, Java, React, LLM orchestration, prompt engineering, cloud computing, relational and NoSQL databases**, deploying scalable applications for complex enterprise workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built scalable FastAPI and LLM orchestration services processing 5,000+ monthly jobs with explicit state transitions, failure isolation, validation, and reconciliation, increasing concurrent throughput 3x while achieving 99%+ reliability.
- Designed multi-tenant PostgreSQL and SQL Server storage for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through normalized schemas, indexing, caching, profiling, and optimization.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Shipped React and Flask applications integrating multimodal AI models, OCR, AWS RDS, and REST APIs, supporting 150+ instructors and efficiently processing 10,000+ daily requests with reliable performance at scale.
- Built prompt chains across GPT-4o, Claude, and Qwen for generation, grading, feedback, and reasoning, adding automated tests, diagnostics, fallback paths, and cost tracking while reducing turnaround time 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Developed Python and PyTorch pipelines processing 100,000+ trajectory records, reliably improving long-horizon planning 18% through rapid prototyping, simulation, debugging, reproducible testing, and iterative model optimization at scale.
- Collaborated with senior researchers to evaluate edge cases and system interactions, increasing closed-loop stability 15% while documenting architecture tradeoffs, experimental findings, technical risks, and reproducible implementation decisions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Java and Python microservices using REST APIs, Docker, SQL, AWS Glue, and S3, improving response times 30%, reducing payload sizes 25%, and lowering storage costs 15%.
- Owned requirements-through-deployment improvements, automating 20+ unit and integration tests across Windows, macOS, and Linux while increasing pipeline efficiency 68% and collaborating with Microsoft and GitHub stakeholders during Agile reviews.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Designed an end-to-end customer solution, mapped 12+ integration points, and presented architecture recommendations to the client before implementing cross-platform booking, payment, notification, and real-time status workflows at scale.
- Built and troubleshot Flask APIs, Firebase synchronization, pricing, coupons, and payment webhooks across 50,000+ records, delivering sub-200-millisecond responses and reducing manual refund handling by 70% for customers.

Skills

Languages: Python, JavaScript, Java, TypeScript, SQL, C++, Shell, Bash
Full Stack/AI: React, Next.js, Node.js, FastAPI, Flask, LLMs, prompt engineering, LLM orchestration, REST APIs
Data/Cloud: PostgreSQL, SQL Server, MongoDB, Redis, AWS, Azure, GCP, Docker, Kubernetes
Engineering: Data structures, algorithms, OOP, Git, CI/CD, unit testing, integration testing, debugging, profiling, optimization

Projects

Big-Brain Live | *Enterprise Generative AI*
- Built AI-powered retrieval over 50,000+ documents using PostgreSQL, GCP, REST APIs, LLaMA-2, FAISS, and Pinecone, achieving sub-second search while reducing retrieval latency 40% through scalable pipeline optimization.

Chat2SQL Code | *Enterprise Data Analysis Tool*
- Developed an AI-powered workspace across Java and TypeScript services, supporting 10+ databases with schema-aware query generation, structured validation, Docker workflows, and reliable cross-platform deployment for technical users.

PiXMod Live | *Scalable Full-Stack AI Application*
- Built a Next.js AI platform integrating CLIP, SAM, Stable Diffusion, and MongoDB, processing 100,000+ image embeddings while delivering semantic search, generative editing, scalable storage, and subscription workflows.